

Axorvian AI/ML Internship



Internship Task1

Introduction to AI/ML and Data Analysis

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1:Dataset Overview:

- Dataset: IPL_Squad_2023_Auction_Dataset
- Source: Kaggle
- Records:568
- Features:8
- Purpose of Dataset: IPL 2023 auction player analysis
- Data Quality: The dataset had missing values in columns like COST IN ₹ (CR.), Cost IN \$ (000), and 2022 Squad. These were cleaned by removing unsold players and filling missing squad values with "New_Player".

2. Analysis performed:

- Load csv file (`pd.read_csv()`)
- Only three columns had missing value, COST IN ₹ (CR.), Cost IN \$ (000), and 2022 Squad. Dataset had missing value in columns .
- Analyze the data types of every columns (`df.dtypes`)
- Missing value Detection using (`df.isnull().sum()`)
- Statistical Summary find using (`df.describe()`)
- Data Cleaning: Remove unsold player (`df = df[df["Team"] != "Unsold"]`)
Missing Value fill in 2020 Squad using (`df['2022 Squad'].fillna("New_Player")`)
 - Find top expensive player using (`df.nlargest()`)

3.Visualization:

- Pie_Chart (Sold Player VS Unsold Player).The purpose of this chart to understand the proportion of player who were sold versus those who remained unsold in the IPL 2023 auction. Sold player (243),Unsold player(325). 57% of players went unsold, while only 43% were sold.
- Bar Chart: Top 4 Most Expensive Players . To identify the highest-paid players in the IPL 2023 auction. Sam Curran was the most expensive player at ₹18.5 Crores. The price difference between the 1st and 4th player is only ₹2.25 Cr.

4.Key Findings / Insights:

After analyzing the IPL 2023 auction data, I found that only 243 players were sold out of 568, which means 57% players went unsold. This shows the auction was very competitive. Sam Curran was the most expensive player at 18.5 Crores. The top 4 expensive players were all-rounders or wicketkeepers, which shows teams like multiskilled players. Most players were bought at low prices between 0 to 0.5 Crores, and very few players crossed 10 Crores. Many players were retained by their previous teams, which means teams trust their existing players. New players were also bought by franchises, showing interest in fresh talent. There was a big gap between base price and final price, for example Sam Curran's base price was 2 Crores but he sold for 18.5 Crores. All-rounders and bowlers were in high demand, while pure batsmen got lower bids. Star players got premium prices, while domestic players were sold at base price. Overall, teams spent heavily on all-rounders because they give balance to the team.

Kaggle_Link:

<https://www.kaggle.com/datasets/harshalkasat/ipl-2023-squad-auction-dataset>